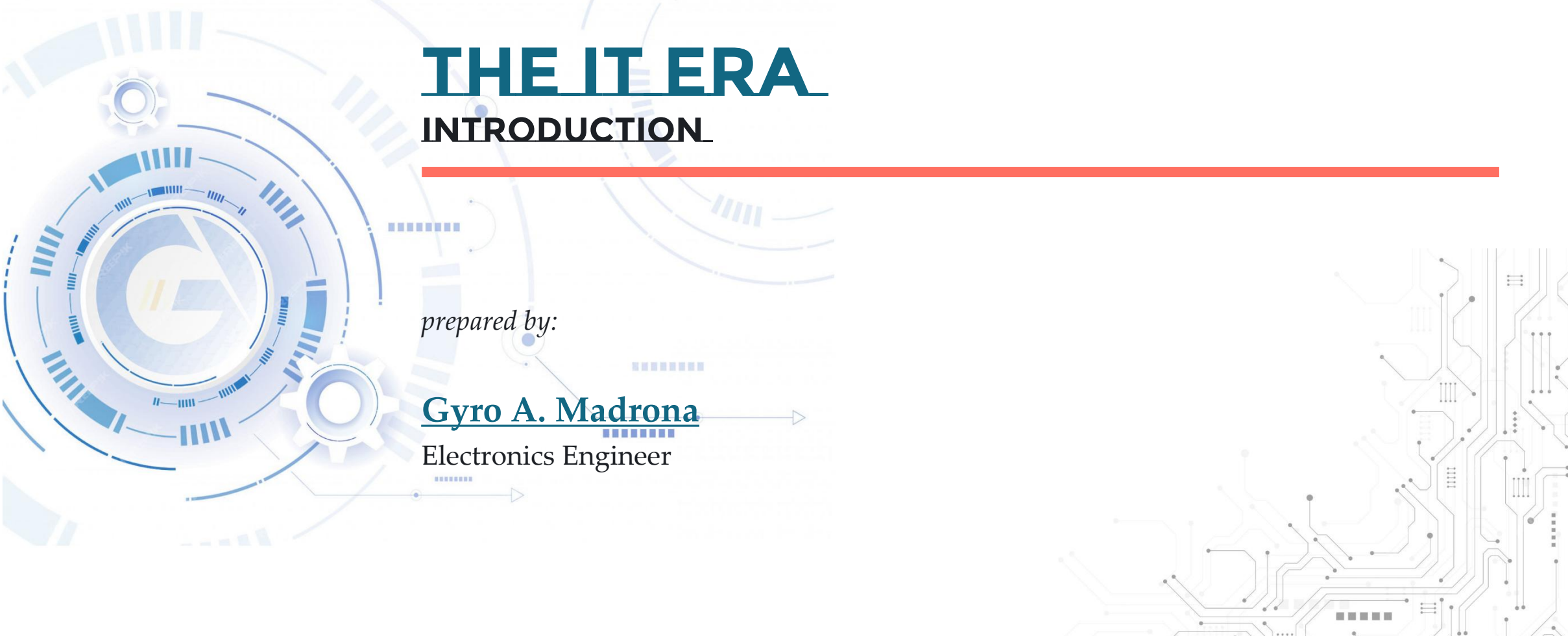




THE IT ERA

INTRODUCTION

prepared by:

Gyro A. Madrona
Electronics Engineer

TOPIC OUTLINE

Information Technology

IT, ICT, Computing

The IT Ecosystems



THINK ABOUT YOUR LAST 24 HOURS

What did you use?

- Smartphone
- Wi-Fi
- GPS
- Digital payments
- Streaming
- Cloud apps
- Sensors
- AI assistants

What enabled it?

- Semiconductors
- Processors
- Software
- Networks
- Data centers
- Satellites
- Algorithms

What change?

- How we communicate
- Work
- Learn
- Travel
- Buy
- Design
- Monitor
- Make decisions



INFORMATION TECHNOLOGY

Information Technology is the use of computing and digital technologies to create, process, store, retrieve, transmit, and manage information.

For engineers, IT is a platform that connects physical systems to computation, communication, control, and decision-making.

Hardware

Physical computing and electronic devices

Software

Instructions and digital services

Data

Digital representation of information



IT, ICT, AND COMPUTING

Information Technology (IT)

Computing technologies used to process and manage information.

Examples: computers, software, storage, databases, cloud services.



IT, ICT, AND COMPUTING

ICT

IT plus technologies for communication and connectivity.

Examples: Internet, mobile networks, telecommunications, collaboration platforms.

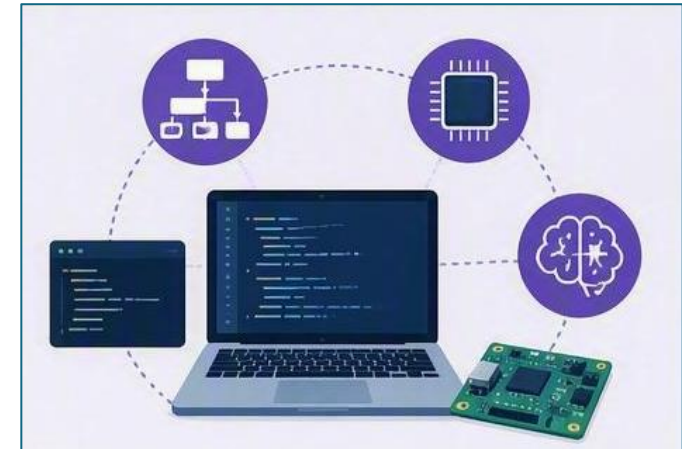


IT, ICT, AND COMPUTING

Computing

The broader discipline of computation and computational systems.

Examples: algorithms, architectures, operating systems, AI, embedded computing.



THE INFORMATION AGE



Fast

Near-instant communication



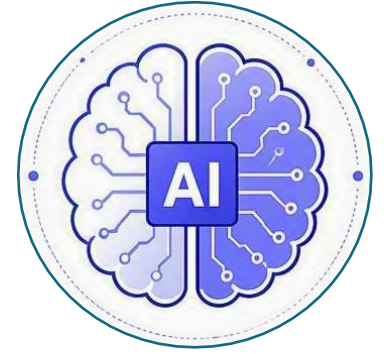
Connected

Always-on networks



Data Driven

Decisions from data



Intelligent

AI-assisted systems



FROM ANALOG TO DIGITAL

ANALOG

Information represented by continuously varying physical quantities.

Examples:

- Vinyl records
- Analog telephone signals
- Film cameras
- Needle gauges

DIGITAL

Information represented using discrete values, typically binary.

Examples:

- Digital audio
- Digital images
- Smartphones
- Digital sensors



THE DIGITAL REVOLUTION

Computing

Mainframes



PCs



Mobile devices

Networking

ARPANET



Internet



Broadband



5G

WEB

Static pages



Platforms



Cloud Services

MOBILE

Smartphones



Apps



Ubiquitous
connectivity

Intelligence

AI



Generative AI



Autonomous
systems



THE IT ECOSYSTEMS

People & Society — Users, organizations, policies, culture

Applications — Apps, websites, enterprise systems, AI tools

Data & Services — Databases, cloud services, analytics

Networks — Internet, Wi-Fi, cellular, fiber, satellites

Computing — CPUs, GPUs, servers, edge devices

Electronics — Semiconductors, sensors, actuators, power



WHY IT MATTERS TO ECE AND COMPUTER ENGINEERING?

Electronics Engineering

Physical foundation: semiconductors, sensors, actuators, interfaces, and communication circuits.

Computer Engineering

Integrates hardware and software to create computational systems.

IT

Information layer: connectivity, data, services, automation, and intelligent decision support.

Modern engineering increasingly operates across the physical–digital boundary.



TECHNOLOGY CONVERGE

Electronics

Semiconductors
Sensors
RF systems
Embedded hardware

Computing

Processors
Operating Systems
Algorithms
AI

Communications

Fiber
Wi-Fi
Cellular
Satellites

Applications

Smart homes
Robotics
Industry 4.0
Smart cities



WHERE DID WE SEE THE IT ERA?

Smartphones

Computing + sensors
+ wireless + cloud

Smart Homes

IoT devices +
automation + apps

Digital Payments

Networks + security +
databases

Telemedicine

Connectivity + data +
digital services

Smart Manufacturing

Robots + sensors +
analytics

Disaster Monitoring

Sensors +
communications + data

AI Assistants

Machine learning +
language + cloud/edge

Autonomous Systems

Sensing + computing
+ control



OPPORTUNITIES AND RISKS

Opportunities

- Faster communication and collaboration
- Better access to information
- Automation and productivity
- New products and services
- Remote work and learning
- Data-driven engineering
- Improved monitoring and safety

Risk and Challenges

- Cybersecurity threats
- Privacy loss and surveillance
- Misinformation and deepfakes
- Digital inequality
- Job disruption
- E-waste and energy use
- Algorithmic bias and overreliance on AI



WHAT IS THE ENGINEER'S ROLE?

Understand

Know how technologies work, interact, and fail.

Design

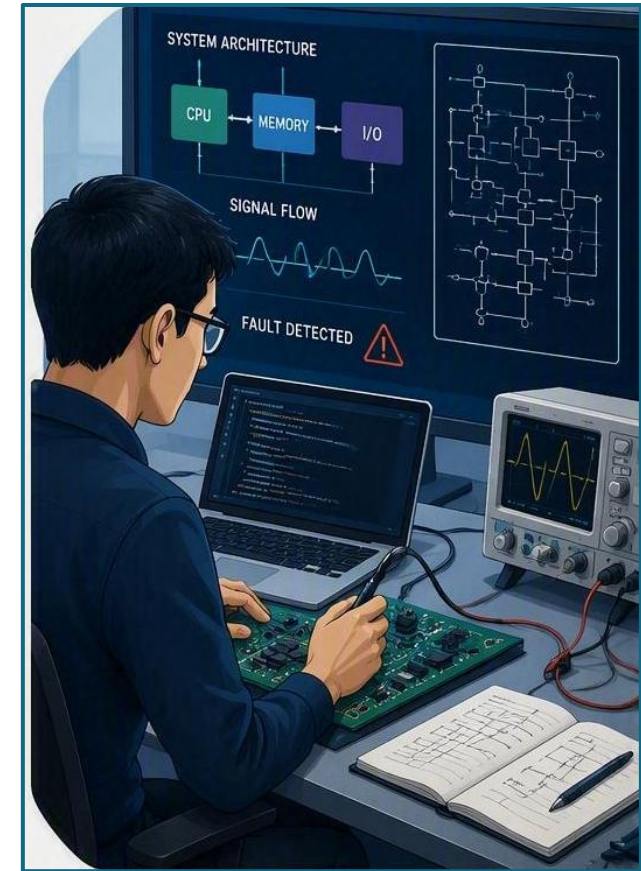
Building systems that are reliable, safe, secure, and useful.

Evaluate

Question performance, cost, risk, ethics, and sustainability.

Respond

Keep learning and adapt as technology changes.



ASSIGNMENT



MY DREAM ENGINEERING CAREER

Presentation Format

Duration: **2 minutes**

Slides : 2 to 3 PowerPoint slides

Include:

- Career Title
- Industry
- Brief description of the role
- Why you choose it
- Required Skills

